

Quantitative Comparison of Manifold-based and Remote-Sensing Clusters for Manufacturing Regions in China: An Interactive Diagnostic Framework

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1. Introduction

Manufacturing regions are spatially structured systems shaped by industrial concentration, infrastructure, logistics and accessibility. China is a strong case study because coastal corridors, inland production belts and major urban clusters express manufacturing intensity differently, meaning analytical conclusions may depend on representation choice [1,2]. The same geography can be described through structural evidence from gridded manufacturing data, mapped infrastructure, and port accessibility [3-5], or through remote-sensing proxies such as VIIRS night-time lights and Sentinel-1 backscatter [6-8]. Combined with dimensionality reduction, density-based clustering, and spatial diagnostics [9-13], these data sources make comparative analysis feasible.

This research matters as analysts increasingly use public industrial and earth observation data to interpret infrastructure accessibility and concentration [1,2]. The same framework could also support applied industrial analytics including supply chain monitoring and market assessment, where direct structural labels are incomplete and proxy signals must be interpreted carefully.

The problem is worth solving now since national structure and remote-sensing data can be assembled over the same spatial units [3,6-8], however conflicting outputs may still produce different research narratives if their concordance (agreement) and instability are not made explicit. With structural and remote-sensing clustering outcomes often interpreted separately, making it difficult to judge whether patterns are genuinely concordant or just superficial. This becomes important when non-linear methods like Uniform Manifold Approximation and Projection (UMAP) remain sensitive to stochastic initialisation and parameter settings [10,14-16].

Current research contains mature methods for dimensionality reduction and density-based clustering [9-11,17], partition comparison and stability assessment [16,18-20], spatial diagnostics [12,13,21], remote-sensing proxies for industrial activity [6-8], and visual analytics for comparative reasoning [14,22-24]. What remains missing is a comparison on a shared grid across structural and remote-sensing representations that uses concordance (agreement), stability, and spatial diagnostics together, with an interpretive layer that explains where agreement, divergence, and instability occur. Figure 1 shows the literature gap addressed by this research, and figure 2 expands the problem logic.

Figure 1: Literature gap diagram

Positioning the research across industrial regions, remote-sensing, clustering, and visual analytics

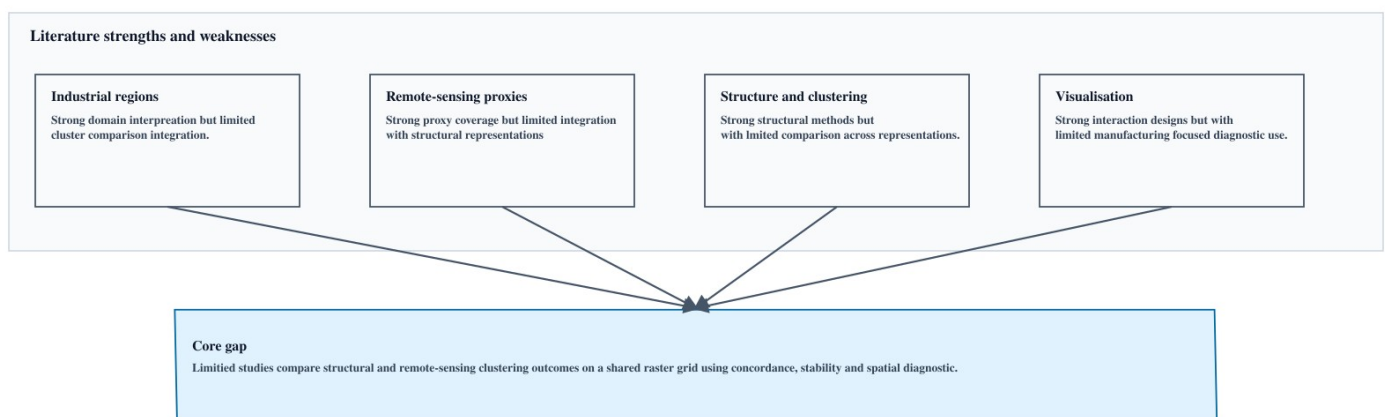
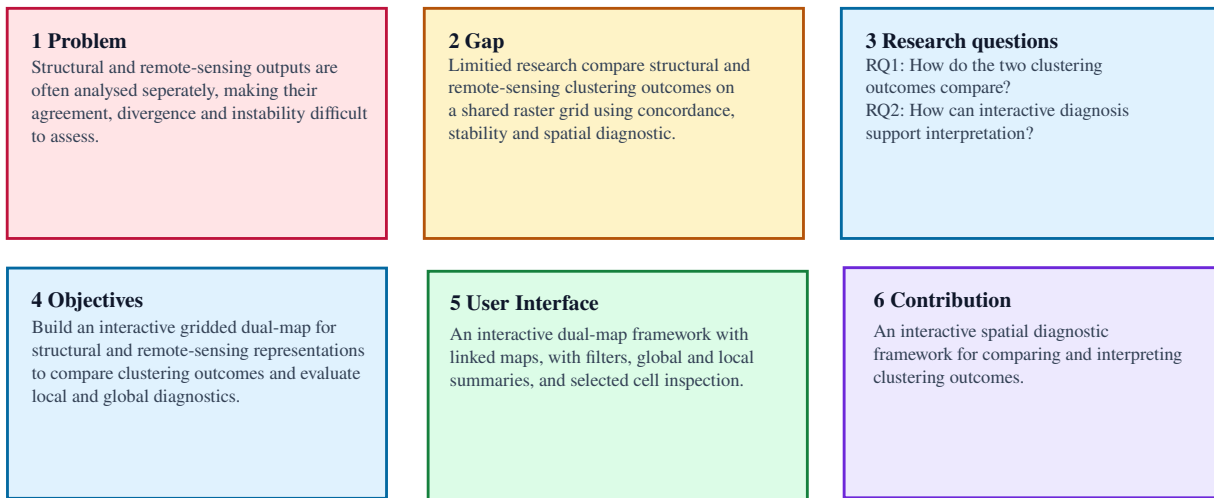


Figure 2: Problem framing and contribution



How can an interactive spatial diagnostic framework support the comparison of concordance , stability and spatial diagnostics between clusters derived from manifold-based and remote-sensing representations on a shared manufacturing raster grid in China? This can be broken down into two sub questions

- RQ1: To what extent do clustering outcomes derived from manifold-based and remote-sensing representations align in terms of concordance, stability and spatial diagnostics on a shared manufacturing raster grid in China?
- RQ2: How can an interactive diagnostic framework support analysts in identifying and interpreting where clustering outcomes agree, diverge or become unstable?

The aim is to compare and interpret clustering outcomes derived from manifold-based structural representations and remote-sensing representations over a shared manufacturing raster grid in China using concordance, stability, and spatial diagnostics, supported by an interactive diagnostic visualisation framework. With the aim broken down into five core objectives.

- O1. Assemble structural, spatial and remote-sensing data from open-source and publicly available repositories, and preprocess the data onto a shared geospatial raster grid.
- O2. Construct manifold-based and remote-sensing representations and use them to generate clustering outcomes within a common clustering framework.
- O3. Quantify within-method stability and cross-representation concordance between manifold-based and remote-sensing clustering outcomes.
- O4. Use global and local spatial diagnostics to assess spatial consistency.
- O5. Design and build a lightweight diagnostic framework to support inspection of where clusters agree, diverge or become unstable.

The objectives show the contributions which this research will provide, including a shared-grid comparison methodology, a quantitative evaluation framework combining concordance measures such as Adjusted Rand Index (ARI) with repeated-run and bootstrap stability, alongside spatial diagnostics using Moran's I and Local Spatial of Spatial Association (LISA) [12,13,18-20], and an interpretive visualisation framework for identifying where clusters agree, diverge or become unstable using a linked dual-map spatial view [21-24].

The research uses public or open-source data and assumes no external labels for manufacturing region types. Remote-sensing measures are treated as indirect industrial proxies rather than definitive ground truth [6-8], and the interface is intentionally scoped as a prototype rather than a full production system. These assumptions narrow the project to a feasible proposal while preserving the central research problem.

2. Background: Remote Sensing and Spatial Analysis for Industrial Areas

2.1 Industrial Areas Patterns in Geospatial Data

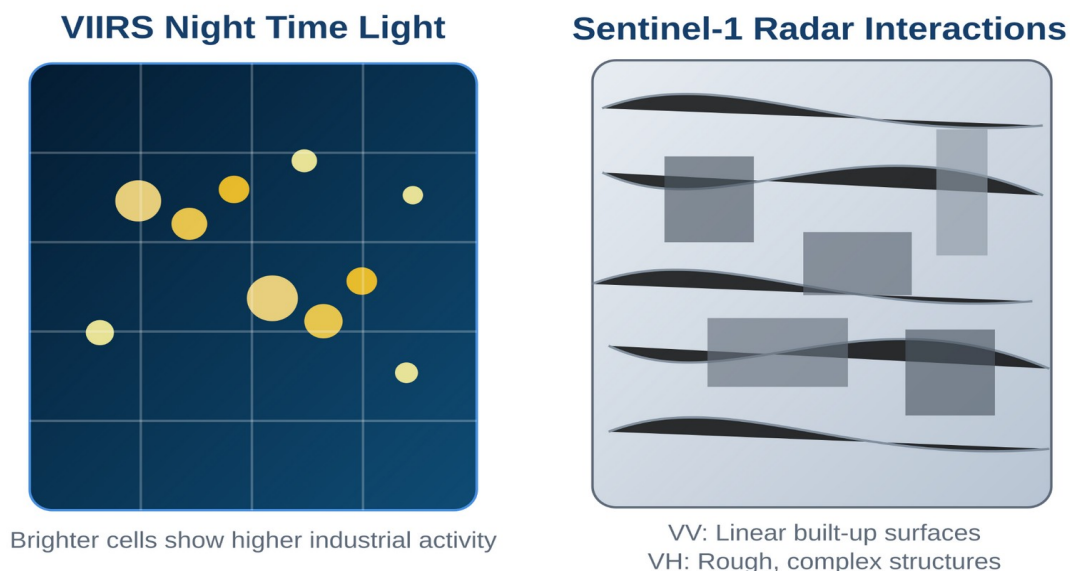
Industrial areas can be observed through direct structural indicators such as manufacturing counts, industrial land, roads, and logistics accessibility, or through remote-sensing proxy signals of illumination, building formations and surface texture. This research treats these as comparable views of the same spatial system rather than interchangeable measures. Structural data may recover latent structures of industrial organisations where as remote-sensing proxy signals may recover broader patterns of intensity or built-up concentration. Analysts therefore need to know whether both views support the same regional interpretation.

2.2 Remote-Sensing as a Proxy for Industrial Activity

VIIRS Night-Time Lights, provides consistent national coverage and captures broad differences in activity intensity, making it useful as a useful remote-sensing proxy for economic activity or densely populated areas of infrastructure [6].

Sentinel-1 Radar Backscatter, uses VV and VH backscatter, these are useful since they capture built-up structure and surface texture associated with industrial land and infrastructure, although they remain indirect because radar signatures do not isolate manufacturing from other dense land-uses [7,8].

Figure 3: Remote-sensing proxies used



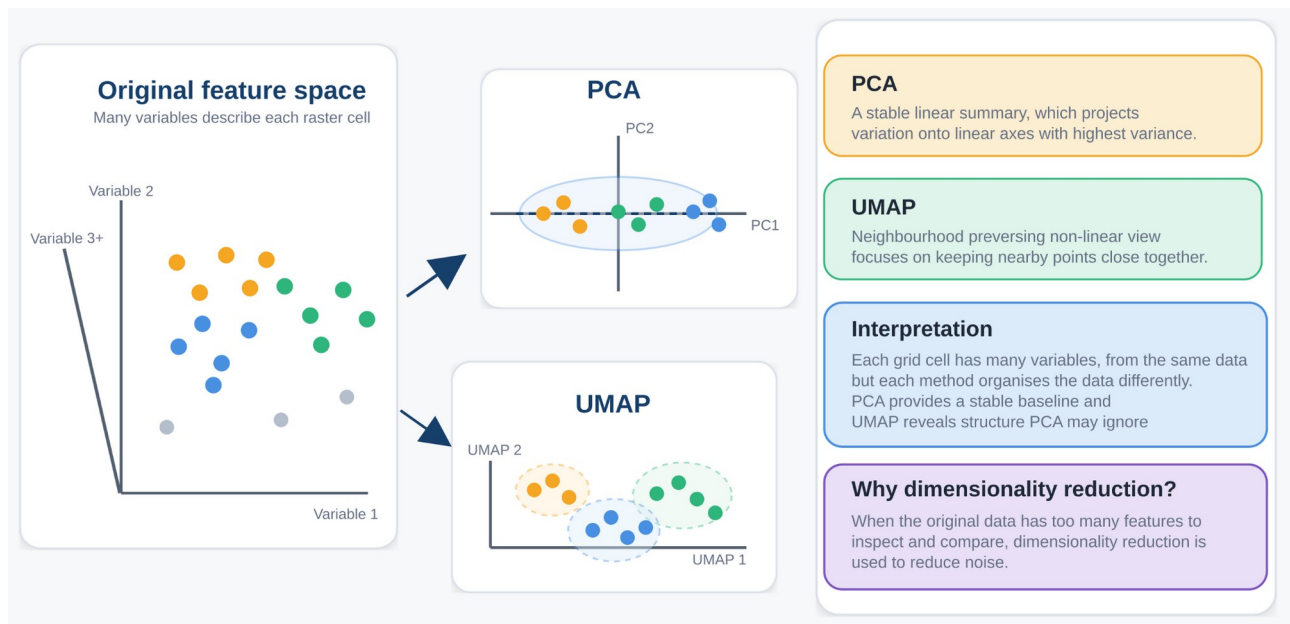
2.3 Structural Indicators of Industrial Activity from Geospatial Data

China Gridded Manufacturing Dataset (CGMD) provides a direct way to model industrial presence [3], Open street maps (OSM) contributes land-use and road context [4], and the World Port Index (WPI) supports measures of logistics accessibility and external connectivity [5]. Together, these datasets form a structural representation that is more interpretable than a purely sensor-based view because its dimensions correspond to observed industrial and infrastructural characteristics. However, this representation should not be treated as a gold standard, since open geospatial coverage varies across space and any reduced feature set necessarily captures only part of the wider industrial system.

2.4 Dimensionality Reduction for Spatially Structured Industrial Data

Principal Component Analysis (PCA) is kept as the stable linear baseline because it is deterministic, variance-maximising, and relatively interpretable, making it suitable for controlled comparison [9]. UMAP is used with PCA since non-linear embedding may reveal neighbourhood structure that a linear project reduces. While UMAP also scales better than t-SNE for larger datasets [10,25], however the reliability of UMAP must be validated through repetition for a fair consistent outcome [14-16].

Figure 4: PCA and UMAP feature mapping



2.5 Density-Based Clustering for Industrial Pattern Discovery

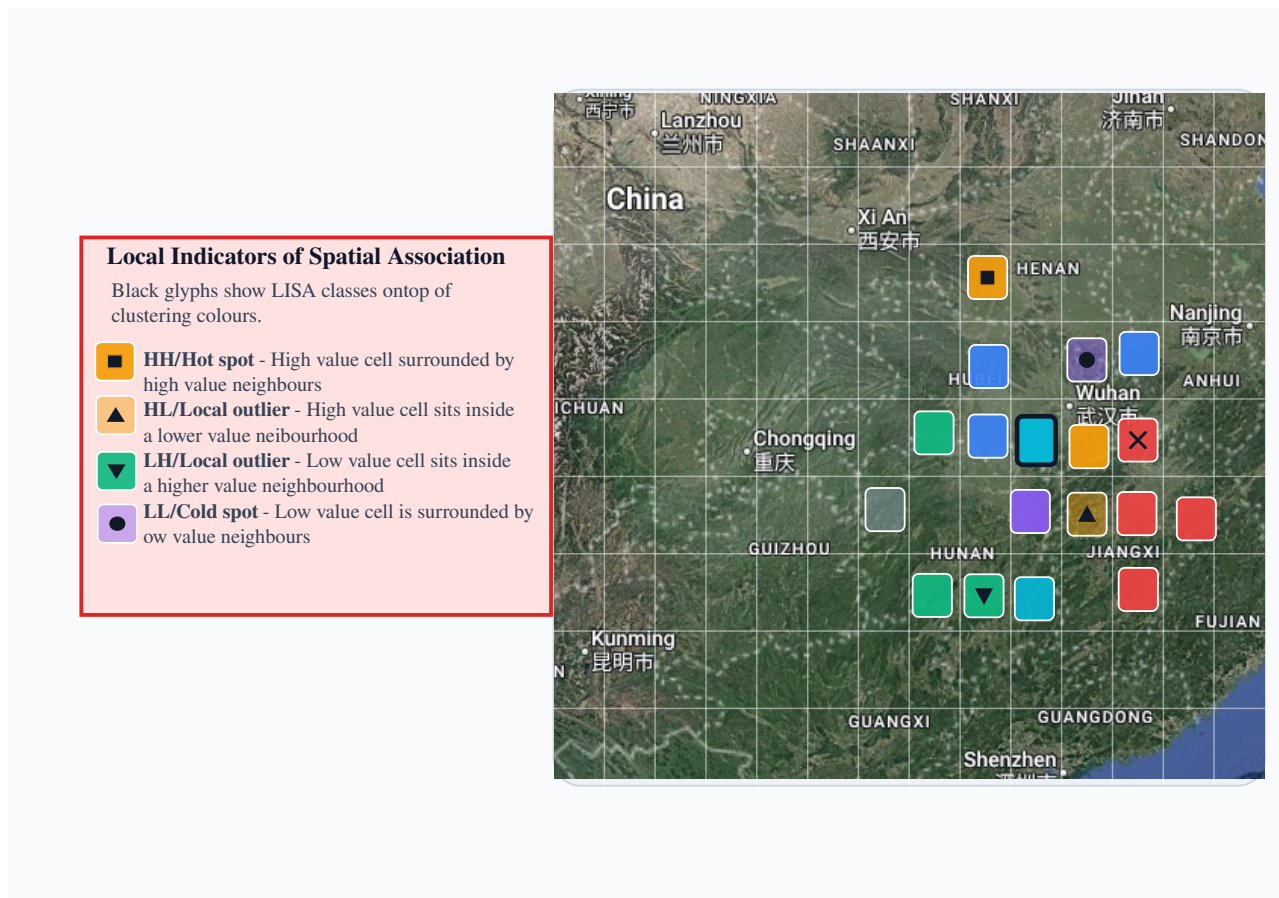
Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) is preferred over k-means and other rigidly shaped clustering methods because industrial space is unlikely to form spherical, equally sized, or fully assigned clusters. A density-based method is more appropriate because it can capture irregular spatial structure while preserving a meaningful noise category [11,17]. Cluster labels are arbitrary across many runs and across representations so direct comparison would be misleading without alignment. Hungarian matching is therefore required before comparison to assure labels are assigned to the best representations, after which ARI can be used as a chance-corrected measure of partition similarity [18].

2.6 Spatial Diagnostics for Clustered Geographic Data

Exploratory Spatial Data Analysis (ESDA) provides the spatial diagnostic layer for this research. ESDA is used to test whether clustering outputs are spatially consistent rather than randomly arranged over a space. Moran's I gives the global measure of spatial autocorrelation, showing which labels are spatially concentrated overall [13]. Local Indicators of Spatial Association (LISA) provides the local ESDA layer, for identifying hotspots, coldspots, and outliers that a single global metric may conceal [12,21]. This is important since the research focuses on global concordance to local exploration, not just where spatial structure exists but where clustering outcomes agree, diverge or become unstable.

Figure 5 shows how LISA encodings will work, the colours are used as illustrative distinctions between structural and remote-sensing representations.

Figure 5: LISA Diagram



3. Related Work on Industrial Area Analysis, Clustering and Diagnostic Visualisation

3.1 Diagnostic and Interactive Visualisation for Cluster Interpretation

Visualisation research is strong at supporting linked comparison, cluster interpretation, and scalable map interaction [21-24,27]. However it cannot in most cases tie those interactions to a comparable method for structure versus remote-sensing representations. Existing studies often support exploration after analysis, whereas this research uses the interface as part of the analytical logic itself.

3.2 Remote-Sensing Based Analysis of Industrial Regions

Existing remote-sensing work is strong at recovering large-scale economic and built-up patterns, and VIIRS combined with Sentinel-1 provide plausible industrial proxies at a global scale [6-8]. However current research still does not reliably separate visually plausible proxy map data for quantitatively based representations of manufacturing structure. This matters since a core challenge is not whether the proxy data is useful but whether the proxy-derived clustering remains comparable with structural representations, over the same cells.

3.3 Dimensionality Reduction and Clustering for Spatially Structured Industrial Data

The dimensionality-reduction research provides strong foundations for linear and non-linear embedding [9,10,25]. Reliability focussed research shows that embedding outputs should be treated as analytical by-products that require validation under uncertainty [14-16]. However, the use of these methods usually stop at the embedding itself, it does not show how instability propagates into downstream clustering, nor how the uncertainty should be interpreted once outputs are mapped.

3.4 Concordance, Stability, and Uncertainty

Partition-comparison and stability research provides the ingredients for evaluating unsupervised outputs without external labels [18-20,26]. Unlike prior work that evaluates reliability within one representation, this research compares structural and proxy-derived partitions over a shared grid. The limitation of existing work is therefore not lack of statistical machinery, but lack of a cross-representation design that keeps agreement, instability, and spatial explanation in the same frame.

3.5 Limitations in Existing Work

Existing research usually stops at a single analytical boundary, remote-sensing is often validated through plausible map readings rather than clustering comparison. Stability research often exposing uncertainty without locating where it matters spatially. Visual analytic research, supports inspection without connecting the concordance, local stability and local spatial diagnostics into a single controlled framework. The result is a fragmented workflow where each component is mature but interpreted independently.

The gap is therefore specific, existing work does not compare structural and remote-sensing clusters on a shared raster grid using concordance, stability and spatial diagnostics with an interpretative visualisation framework, Figure 6 explicitly shows this.

Figure 6: Literature positioning matrix

Paper	R1 Shared-grid comparison	R2 Global diagnostics	R3 Local diagnostics	R4 Spatial and attribute interpretation
Fan C, et al., China's Gridded Manufacturing Dataset. Scientific Data, 2022.	✓			
Elvidge CD, et al., VIIRS Night-Time Lights. International Journal of Remote Sensing, 2017.	✓			
Chini M, et al., Global Building Map from Sentinel-1 SAR Data. Remote Sensing, 2018.	✓			
Jolliffe IT, et al., Principal Component Analysis. Philosophical Transactions of the Royal Society A, 2016.	✓			
McInnes L, et al., UMAP for Dimension Reduction. arXiv, 2018.	✓			
Malzer C, et al., Hybrid Density-Based Cluster Selection. IEEE MFI, 2020.	✓			
Steinley D, et al., Co-Clustering Adjusted Rand Index. Journal of Classification, 2020.		✓		
Yu H, et al., Bootstrapping Estimates of Stability for Clusters. Computational Statistics, 2018.		✓		
Hennig C., Cluster-wise Assessment of Cluster Stability. Computational Statistics & Data Analysis, 2007.		✓	✓	
Chen Y., Moran's Index of Spatial Autocorrelation. arXiv, 2016.		✓		
Anselin L., Local Indicators of Spatial Association (LISA). Geographical Analysis, 1995.			✓	✓
Mason L, et al., Demystifying Spatial Dependence. IEEE VIS, 2024.			✓	✓
Roberts JC., Coordinated & Multiple Views in Exploratory Visualization. CMV, 2007.				✓
Sedlmair M, et al., Visual Cluster Separation Factors. Computer Graphics Forum, 2012.		✓		✓
Jung M, et al., GhostUMAP. IEEE VIS, 2024.		✓	✓	
This research	✓	✓	✓	✓

4. Data Sources, Shared Raster Grid and Preprocessing

4.1 Spatial Framework and Shared Raster Grid

All datasets are aggregated onto a shared raster grid covering China, as a common geospatial case study covering, exploring representation of preprocessed-data, clustering, and evaluation. This is not a convenience choice, it's the methodological condition that makes cross-representation comparison valid at both global and local scales.

4.2 Structural and Remote-Sensing Data Sources

The structural representation combines CGMD [3], industrial land-use and road context from OSM [4], and logistics accessibility from the World Port Index [5]. The remote-sensing representation combines VIIRS with Sentinel-1 VV and VH summaries [6-8]. Together, these sources provide two unique views of the industrial space. One for structural variables and the other as a continuous proxy signal for national coverage. Concordance between them strengthens comparative confidence, while divergence becomes a diagnostic problem rather than a simple modelling error.

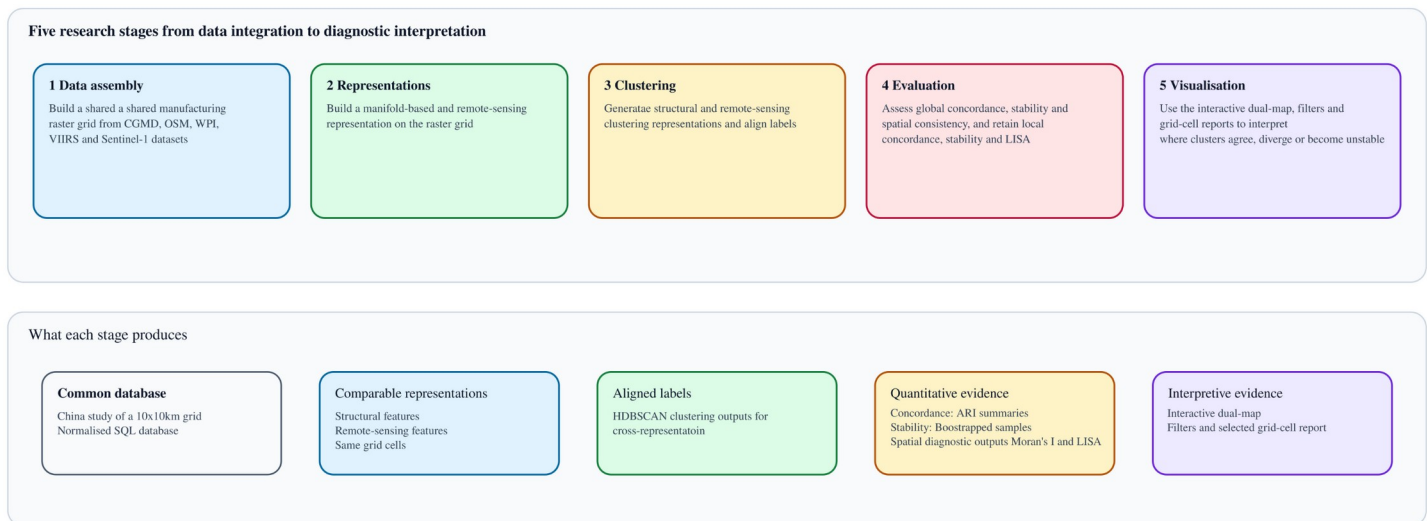
4.3 Preprocessing, Standardisation, and Multi-Source Integration

Variables from different sources are not directly comparable as counts, densities, distances and sensor-derived metrics measure differently across scale, distribution and interpretation. This is where preprocessing standardises and cleans missing values, and aggregates data onto a shared grid. These derived variables such as manufacturer density and weighted port distance are retained because they show industrial structure more clearly than raw counts alone.

4.4 Data Constraints and Absence of Ground Truth

This research does not assume reliable external labels for manufacturing regions, so it cannot be evaluated against a known partition [26]. The absence of ground truth is a methodological constraint but is justified by adopting a comparative design. Public geospatial data is uneven and remote-sensing variables remain indirect measures of manufacturing activity rather than direct observations [4,6-8]. Therefore the contribution is to ensure representational agreement, instability and spatial concentration is explicitly measurable and interpretable since no single representation can claim truth.

Figure 7: Methodology pipeline overview



5. Methodology and Diagnostic Visualisation Framework

5.1 Analytical Framework Overview

This research compares methodology in which structural and remote-sensing representations are constructed on the same grid, clustered independently, aligned for comparison, evaluated quantitatively then interpreted through global and local spatial diagnostics. Figure 7 makes this sequence explicit, while Figure 6 shows how the results will be read within a diagnostic prototype. The originality is not from proposing a new clustering algorithm, but in integrating representation construction, comparison, stability assessment, and interpretation within one evidential framework.

5.2 Representation Construction and Feature Spaces

The structural representation feature space combines manufacturing presence, industrial land-use, road context, and logistics accessibility from CGMD, OSM, and world port data, as these variables remain directly interpretable [3-5]. Dimensionality reduction methods PCA and UMAP are both retained as research needs a deterministic linear baseline and a non-linear alternative. Using only PCA would risk suppressing local structure while only using UMAP would remove the stable comparison required for controlled interpretation. [9,10,14-16]. Remote-sensing representation as a proxy feature space is built from VIIRS and Sentinel-1 VV and VH summaries, this is treated as a competing representation rather than an inferior validation layer [6-8].

5.3 Density-Based Clustering and Cross-Representation Comparison

HDBSCAN is used since industrial regions are unlikely to form clusters that are spherical or equally sized, making a density based method appropriate since it handles noise without requiring a fixed cluster count [11,17]. Each representation then produces a partition over the same grid-cells so that comparison remains direct, with the Hungarian matching algorithm required before comparison to avoid misleading label differences. ARI is then used as a probability corrected measure of partition similarity, which is suitable for comprising unsupervised partitions where no reliable external labels exist [18].

5.4 Stability Assessment and Spatial Consistency

UMAP-based clustering stability is by repeatedly running across multiple seeds to estimate the sensitivity to stochastic initialisation [10,15]. Bootstrap resampling is used to test whether clustering structure remains consistent under data distortion [16,19,20]. Pairwise ARI across repeated runs then summarises within-method stability. Global spatial consistency is assessed with Moran's I, which tests whether cluster labels show consistent spatial concentrations rather than random geographic scattering [13].

LISA complements this by identifying local hotspots, coldspots, and outliers, while local concordance matches cells and local stability to show how often cells remain within its matched clusters [12,21].

5.5 Diagnostic Framework Design

The diagnostic visualisation framework is structured around three diagnostic tasks: grid-level comparison, cluster interpretation, and attribute-based spatial reasoning. Figure 6 integrates these tasks through filters that define the analytic subset, linked maps that provide matched spatial context, summary panels that convert comparison into global evidence and a selected cell report that turns a location into an interpretable case [21-23]. Task1 supports grid-cell cross representation comparison by showing how individual grid cells are assigned across dimensionality-reduction and remote-sensing cluster maps. Task2 supports cluster interpretation and stability inspection by revealing cluster footprints, summaries, concordance and instability

across runs and representations. Task3 test whether agreement, divergence or instability is associated with variables such as density, weighted distance, VIIRS, Sentinel-1, LISA class, local concordance, and local stability [24, 27].

Figures 8-10 show how the requirements are met, and the components of the user interface, along with diagnostic tasks and core interactions. The colours, numbers and labels are used as illustrative distinctions between structural and remote-sensing representations. The cluster encoding key is also illustrative.

Requirement	User interface component
R1: Shared-grid comparison: Support direct comparison of the same grid cell across representations on a common raster grid.	U1: Linked dual-maps: Compare structural and remote-sensing cluster assignment on the same raster grid.
R2: Spatial and attribute interpretation: Be able to explain local patterns using spatial diagnostics and attribute context.	U2: Filter Panel: Filter attributes, local diagnostics, and LISA classes. U4: Clicked cell report: Show local metrics and attributes for selected cell. U5: Interpretation guide: Show cluster encodings, LISA encodings and interaction meanings.
R3: Global diagnostics: Summary metrics for concordance, stability and spatial consistency across representations.	U3: Global Summaries: Show concordance, stability and spatial consistency.
R4: Local diagnostics: Support local inspection where representations agree, diverge or become unstable.	U1, U2, U4
Diagnostic task	Core interaction
T1: Grid-Level Cross-Representation Comparison: Compare cluster assignments across representations at the grid-cell level.	I1: Linked dual-maps select and compare grid cells across representations.
T2: Cluster Interpretation and Stability Inspection: Interpret what clusters represent and assess stability across runs and representations.	I2: Highlight, summarise and inspect clusters.
T3: Spatial Reasoning with Attributes: Examine whether specific grid cells agree, diverge or become unstable with associated specific attributes.	I3: Filters by attributes and local diagnostics.

Figure 8: Requirements



Figure 9: User Interface components



Figure 10: Diagnostic Tasks And Core Interactions



5.6 Expected Outputs And Interpretation

Figures 8-10, show how the methodology works as a full diagnostic interface. Figure 8 shows the system requirements, figure 9 shows the main interface components and figure 10 defines the diagnostic tasks and interactions supported by the design. Combined they show how quantitative outputs will be interpretable rather than just displayed.

The design of the interface combines global and local evidence into a single interactive view, supporting non-representation inspection, summary panels, selected cell report, and concordance, stability and spatial diagnostics. This linked dual-map provides an interpretive layer through quantitative outputs rather than a full production system. These figures show that this is not a simple dimensionality reduction or clustering exercise, but a thoughtful comparative framework for explaining where representations agree, diverge, or become unstable.

5.7 Implementation Overview and Tools

Python provides an excellent reproducible environment for data handling, machine learning, spatial statistics, and visual output. sqlite3 supports a lightweight raster grid database, pandas and geopandas support data integration, scikit-learn supports preprocessing and PCA, umap-learn and hdbscan support the main representations and clustering methods, while libpysal and esda support Moran's I and LISA for exploratory spatial data analysis (ESDA) [9-12,24]. Matplotlib and Kepler.gl provide a lightweight WebGL web-mapping stack which supports figure generation, interactive views and selected cell reporting [24].

6. Evaluation Design

6.1 Evaluation of RQ1 (Clustering Comparison)

RQ1 is evaluated through evidence at three levels: concordance, stability, and spatial consistency. ARI is used to measure how well structural and remote-sensing clusters align on the same shared raster grid, repeated runs and bootstrap resampling test whether those outcomes are robust and stable. Moran's I and LISA assess whether the resulting partitions show spatial concentration rather than random geographic scattering [12,13,18-20]. Strong results do not require perfect agreement, but interpretable patterns showing where both representations agree, diverge or become unstable.

6.2 Evaluation of RQ2 (Diagnostic Interpretation)

RQ2 will be evaluated through two diagnostic cases. Starting with global metrics, identifying local regions of concordance or divergence, inspection of selected cells and explaining patterns using attributes such as manufacturing density, port accessibility, VIIRS intensity, Sentinel-1 backscatter, LISA class, and local stability.

6.3 Evaluation Outputs and Evidence Criteria

The expected outputs include aligned partitions, ARI summary tables, repeated-run and bootstrap stability results, Moran's I values, LISA layers, local concordance and stability metrics with the interactive prototype itself. These outputs form a single evidential chain, where later interpretation depends on earlier quantitative results. Figure 11 summaries the evidence success criteria that connect quantitative outputs to an interactive visualisation.

6.4 How the Evidence Answers the Research Questions

These outputs answer RQ1 by showing whether structural and remote-sensing clusterings are concordant, robust, and spatially consistent on the same grid. They answer RQ2 by showing whether those outcomes can be interpreted in a disciplined way once local diagnostics and linked visual explanation are available. The evaluation logic is therefore comparative and evidential, rather than descriptive.

Evidence	Requirements supported	Evidence produced
E1: Delivery: The dual-map diagnostic framework is implemented with linked maps, filters, summaries and selected cell report.	R1, R2, R3, R4	Kepler.gl prototype with linked dual-maps, filter panel, global summaries and clicked cell report.
E2: Interpretation: At least 2 examples show how the framework explains concordance, divergence and instability.	R2, R4	Two worked examples showing how the framework explains agreement, divergence and instability.
E3: Quantitative: ARI, repeated run stability, bootstrap stability, Moran's I and LISA outputs are produced.	R3, R4	Quantitative outputs for concordance, stability and spatial consistency.
E4: Task: System supports 3 diagnostic interactions of compare, filter and inspect.	R1, R2, R4	Example showing the system supports the 3 interactions.

Figure 11: Evidence of Requirements



Success of RQ1: Shows strong concordance, stability and spatial consistency across representations on the same grid.

Success of RQ2: Explain a case where global concordance looks acceptable but local divergence or instability reveals otherwise.

7. Limitations, Risks, and Feasibility

7.1 Methodological and Interpretive Limitations

This research assumes no reliable external labels for manufacturing region types, so it cannot be evaluated against a known partition [26,28]. Public geospatial coverage is uneven, and logistics proxies only capture a sub-section of the wider supply-chain structure [4,5]. VIIRS and Sentinel-1 also remain indirect indicators of industrial activity rather than direct observations of manufacturing function [6-8]. These limits do not weaken the design, but they do justify why comparative evaluation is methodologically necessary.

UMAP remains sensitive to stochastic initialisation, and HDBSCAN remains sensitive to the structure presented to it, therefore robustness cannot be assumed from a single visually plausible run[10,11,15,16]. Structural and remote-sensing feature spaces are comparable, but they do not encode congruent clustering labels, so disagreement should not automatically be treated as error. Moran's I and LISA improve interpretation, but they do not remove ambiguity, especially at boundaries between manufacturing regions where local classes depend on neighbourhood definition [12,21].

7.2 Technical and Project Risks

The main technical risks come from data integration, interface delivery, and project timing. Multi-source aggregation may be slowed by cleaning and validation issues across multiple public and open-source datasets. Interactive mapping may also take longer than expected once linked behaviour, filtering, and selected-cell reporting are implemented. The main schedule pressure is therefore likely to come from the overlap between analysis, visualisation, and writing the results of the research.

The project remains feasible because it uses public data, established methods, and a clearly bounded scope. Risks are reduced through staged delivery, reusable code modules, early figure design, and prioritisation of the comparative pipeline before interface refinement.

7.3 Risk Register and Timeline

Figure 12 summarises the risk register and probability-impact matrix. It is useful not only as project-management documentation, but as evidence that the main proposal threats are already understood in relation to scope. With the highest risk coming from the concerns of late stage visualisation and writing rather than fundamental access to data or methods.

Figure 12: Risk assessment chart

ID	Risk	Category	Probability (1-5)	Impact (1-5)	Score	Level
1	Review literature and concept framing becomes too broad	Research	2	4	8	Medium
2	Data integration is delayed or source coverage is incomplete for the shared raster grid	Data	2	4	8	Medium
3	Coding or preprocessing introduces bugs which effect the outputs	Technical	1	3	3	Low
4	Methods require further justification after feedback and review	Method	3	2	6	Low
5	Results are weaker or less consistent than initially expected	Results	3	4	12	Medium
6	Visualisation and mapping take longer than expected	Results	5	4	20	High
7	Write-up in July takes longer than expected	Writing	4	5	20	High
8	Referencing, formatting or chapter integration issues appear late	Writing	2	5	10	Medium

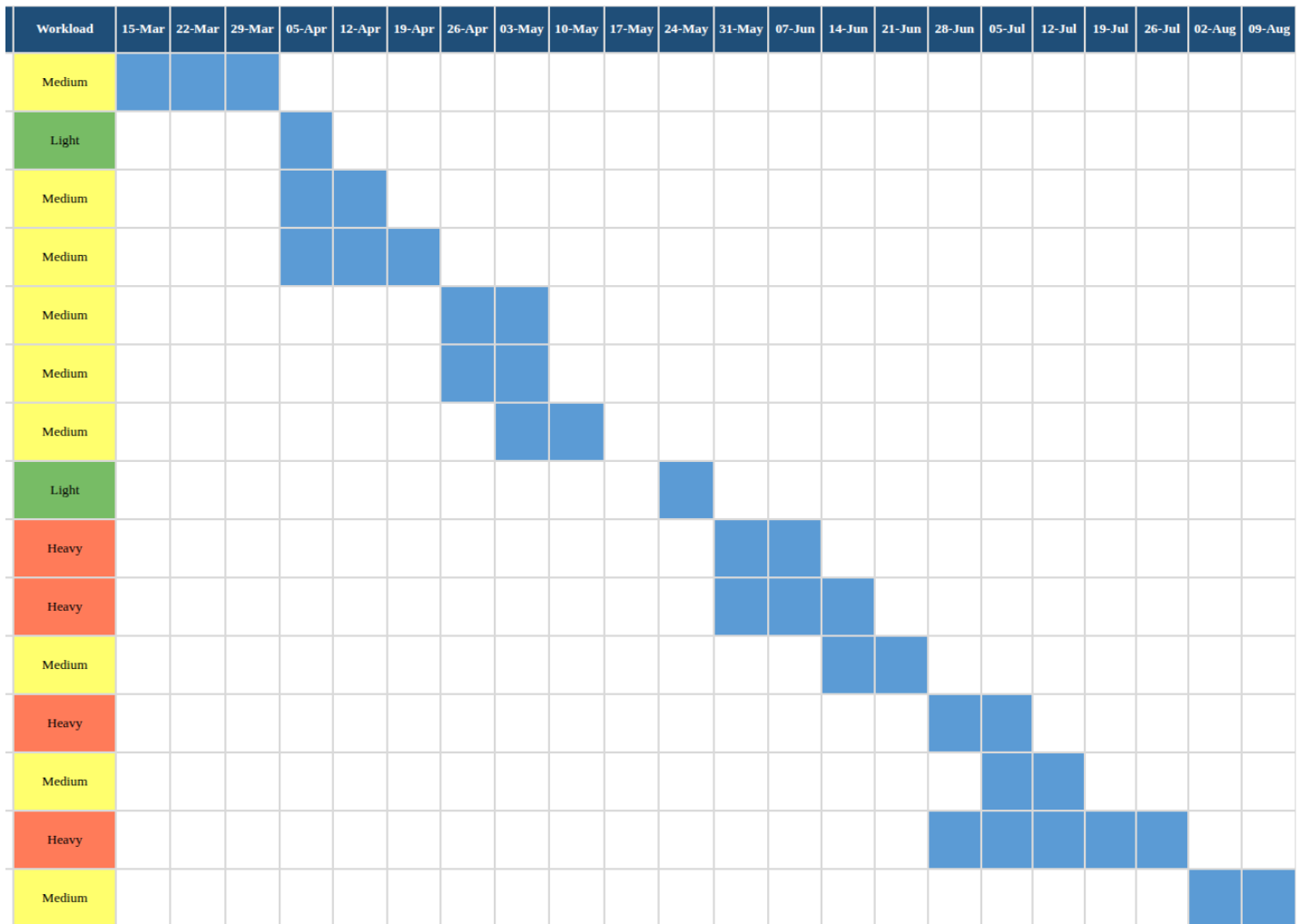
Mitigation	Contingency	Initial Source
Use clear boundaries and weekly reviews	Reduce the view to the most relevant and important points and sources	Researcher
Start extraction early, track completeness and validate at intervals	Reduce scope and datasets used and adjust documentation	Online
Use version control, modular functions and testing stages	Roll back to the last stable version	Researcher
Schedule regular meetings for feedback and review to understand terminology within the literature	Simplify the approach and justify choices explicitly	Reviewer
Run robustness checks and keep all previous versions stored for rollback	Reframe findings and contribution and acknowledge the limitation explicitly	Researcher
Create reusable templates and prioritise the figure list early	Produce only the core figures and draft extras in appendices	Researcher
Lock in chapter design early, draft simultaneously while coding	Prioritise core chapters and reduce non-essential extras	Researcher
Maintain all information and versions used throughout the project and allow enough time to triple check.	Allow enough time prior to deadline for review and adjustments	Researcher

RISK CHART	Impact 1	Impact 2	Impact 3	Impact 4	Impact 5
Probability 5		1		1	
Probability 4		2	1		1
Probability 3	1				
Probability 2			1		
Probability 1					

Figure 13 shows expected timeline from the literature review all the way through to the final writing of the results and a buffer window for any revisions required. It provides administrative value by showing a comparative pipeline schedule before the interactive diagnostic framework is fully implemented, this reduces the chance of delays in result generation or core diagnostic visualisations not being able to be produced.

Figure 13: Project timeline (Gantt chart)

Phase	Task	Start	End	Days	Workload
Planning	Design pipeline, file and database structure	16-Mar-26	04-Apr-26	20	Medium
Data Assembly	Build shared raster grid and output tables	05-Apr-26	11-Apr-26	7	Light
Data Assembly	Collect, preprocess and check CGMD data	05-Apr-26	18-Apr-26	14	Medium
Feature Build	Build OSM and WPI features set	05-Apr-26	25-Apr-26	21	Medium
Feature Build	Process VIIRS features	26-Apr-26	09-May-26	14	Medium
Feature Build	Process Sentinel-1 VV and VH features	26-Apr-26	09-May-26	14	Medium
Data Integration	Normalise and combine all data	03-May-26	16-May-26	14	Medium
Modelling	Build PCA representation	24-May-26	30-May-26	7	Light
Modelling	Build UMAP representation	31-May-26	13-Jun-26	14	Heavy
Clustering	Run HDBSCAN and align labels	31-May-26	20-Jun-26	21	Heavy
Testing	Test repeatable runs and export core outputs	14-Jun-26	27-Jun-26	14	Medium
Evaluation	Compute ARI, stability and concordance outputs	28-Jun-26	11-Jul-26	14	Heavy
Evaluation	Compute Moran's I, LISA, local concordance and local stability	05-Jul-26	18-Jul-26	14	Medium
Interface	Build dual-map, filters and clicked-cell report	28-Jun-26	01-Aug-26	35	Heavy
Final Outputs	Produce final maps, figures and outputs	02-Aug-26	15-Aug-26	14	Medium



8. Conclusion and Future Work

This research proposes a shared-grid diagnostic framework for comparing structural and remote-sensing clusterings of manufacturing regions in China. Rather than treating the absence of ground truth as a weakness, we use it as the rationale for comparative and uncertainty-aware evaluation, to enable decisions under uncertainty.

There are three core contributions: A shared-grid comparative methodology, a quantitative evidence framework for concordance, stability, and spatial diagnostics, and an interpretive visualisation framework for locating agreement, divergence, and instability. These contributions are beneficial as they turn a known clustering problem into a defensible comparative research method. The value of this lies not only in producing clusters but showing how those clusters can be judged and interpreted by analysts. As an ambitious problem, each methodological component has to be realistic, bounded, justified against alternatives and linked directly together.

Future work could extend the framework to other jurisdictions, richer remote-sensing features, analyst evaluation studies, and adjacent industrial systems that face similar comparison problems. It could also

implement additional structural indicators with more advanced diagnostic interactions. Even without predefined labels, the framework would still matter as it provides a reproducible way to compare, interpret and diagnose competing spatial clusterings in a data rich but information poor setting.

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